PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

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Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

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# Benefits of Gamma Aminobutyric Acid (GABA) as Functional Feed Additive in Animals

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# Key Topics

- Introduction about Alternative of Antimicrobials (ATA)
- GABA general information
- GABA benefit for animals & field data

# Introduction about ATA

# Alternative of Antimicrobials (ATA) as functional feed additive

- In order to reduce Antimicrobial Use (AMU) and Antimicrobial Resistance (AMR), the alternative to antimicrobials (ATA) are developed for their benefits to microbiomes and gut health
- Several studies illustrated that ATA may have more advantages than the benefit to animal gut health
  - As antimicrobial alternative
  - As non-Antimicrobial growth promoter
  - As environmental control
  - As reproductive support

# The key characterization of ATA

The parameters need to be concerned before choosing any ATA

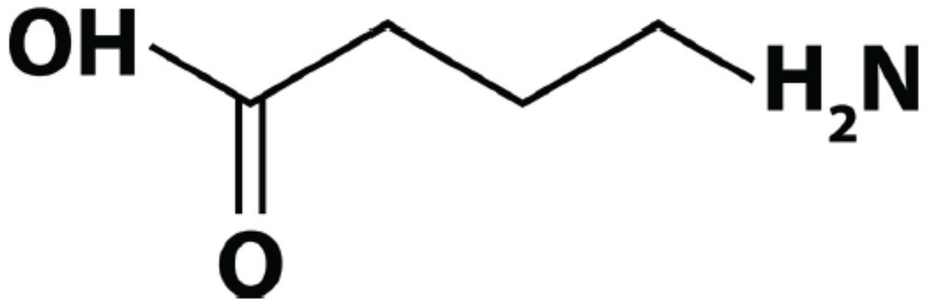
**ATA**

- Physiochemical characterization
  - formulation
  - dosage
  - administration
  - duration
- Functional characterization
  - bioavailability
  - pathway
- Safety evaluation
  - in animals
  - in consumers
  - in environment

such as AMR, toxicity, side effect to carcass quality, etc.

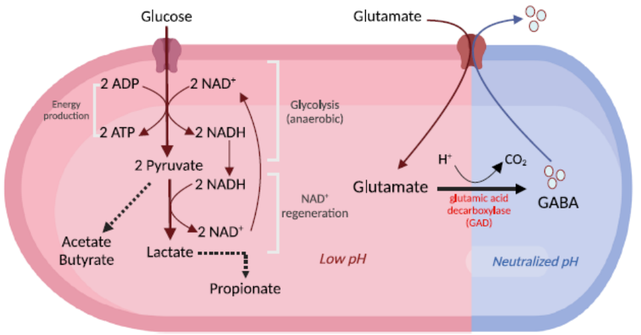
Rahmann et.al., 2022, Antibiotics, 11,766

# GABA general information



# General Information

- Gamma Aminobutyric Acid or GABA is a non-protein amino acid with four carbons
- GABA, naturally, synthesize from decarboxylation of glutamic acid from gut microbiotas
- GABA is an important postbiotic with the relation to others postbiotics such as SCFA
- GABA is an inhibitory neurotransmitter in the CNS of vertebrates and invertebrate phyla including arthropods, echinoderms, and platyhelminths
- GABA has an important role in the animal central thermoregulation including modulating of heat stress
- GABA is not only the  inhibitory neurotransmitter in mammals, but GABA also found in other tissues such as ovary, placenta, and uterus




Barido et al., 2021, Biosci., 34 (4):621-632
Braga et al., 2024, npj Science of food, 8:16, https://doi.org/10.1038/s41538-024-00253-2
Chen et al., 2020, Food Funct., 11, 9067-9074
Liu et.al., 2022, Animals, 12, 2539.
Oketch-Rabah et al., 2021, USP Safety Review of GABA, Nutrients, 13, 2742
Westphalen, D., 2019, Healthline Medical Network
Zhao et al., 2020, Hindawi Mediators of Inflammation, V.2020, Article ID 7368483

# GABA-producing microorganisms

**Table 1.** GABA production using solid-state fermentation with different microorganism species.

| Microorganism | Fermentation Condition | GABA Production (mg/L) | References |
|---|---|---|---|
| *L. brevis RK03* | 30 °C, pH: 4.5, Inoculum: $1 \times 10^{9}$ CFU/mL, 88 h | 62.523 | [41] |
| *S. thermophilus* | 37 °C, pH: 6.0, 48 h | 8300 | [41] |
| *L. brevis* | 37 °C, 48 h | 4 | [42] |
| *L. brevis* | 37 °C, pH: 5.0 | 205 | [43] |
| *L. brevis* | 37 °C, pH: 5.0 | 210 | [43] |
| *L. plantarum FNCC 260* | 37 °C, pH: 5.0 | 352 | [30] |
| *L. brevis HY1* | 30 °C, 48 hr, pH: 5.0 | 18.76 | [44] |
| *L. plantarum K16 strain* | 34 °C, pH 5.5, 96 h | 1284 | [45] |
| *Debaryomyces hansenii* | 29 °C for 24–48 h | 6790 | [46] |
| *L. plantarum K154* | 7 days at 25 °C | 15.53 | [47] |
| *L. plantarum FRT7* | 40 °C, pH 7.0, 48 h | 1158.6 ± 21.22 | [48] |
| *K. marxianus JMY140K* | 40 °C for 48 h | 125.6 | [49] |
| *Debaryomyces hansenii* | 30 hr at 30 °C | 179.2 | [50] |
| *S. cerevisiae DL6–20* | 28 °C, pH 7 | 1890 | [51] |
| *Aspergillus oryzae NSK* | 7 days, 30 °C | 354.08 | [52] |
| *Aspergillus oryzae* | 30 °C, pH 6.0 | 51.53 | [53] |
| *Rhizopus oryzae* | 30 °C, pH 6.0 | 50 | [43] |
| *L. futsaii CS3* | pH 5, 37 °C | 6.840 | [54] |

Ahmad et al., 2025, Fermentations, 11, 84, https://doi.org/10.3390/fermentation11020084

# Mechanism of GABA in microorganisms

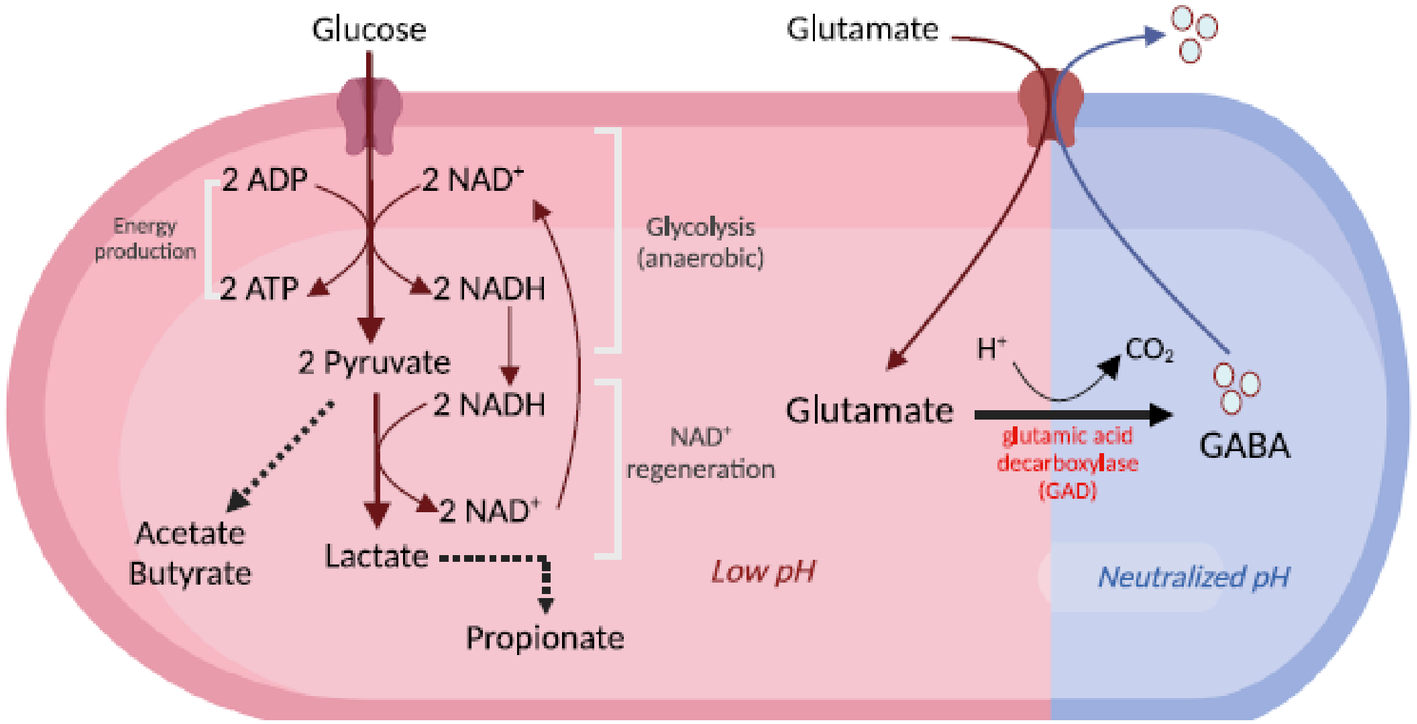



**Fig. 3 | Mechanism of GABA production in microorganisms.** Under anaerobic and acidic conditions in the human gut and fermentation, it appears that bacteria produce GABA for their own survival purposes under these extreme environments. Under anaerobic conditions, glycolysis takes place in the cytosol, where $NAD^+$ and ADP are required to convert glucose into pyruvate, in which NADH and ATP are produced from the process[75]. Pyruvate is then converted into lactate or other organic compounds, such as acetate, butyrate, and propionate, where NADH is utilized, and $NAD^+$ is generated in the process. Then, $NAD^+$ is fed back and reutilized in the glycolysis process[75]. The acidic fermentation by-products, lactate, and other organic compounds lower the pH, which leads bacteria to utilize the GAD gene system and triggers the production of GABA[76,77]. To produce GABA, exogenous glutamate is transported into the cell by a glutamate/GABA antiporter, then glutamate is converted into GABA by glutamic acid decarboxylase (GAD)[36,76,78]. Then, GABA is exported from the cell via the antiporter, resulting in an increase in the pH of the cytoplasm due to the removal of $H^+$ ions and a slight increase in the extracellular pH due to the exchange of extracellular glutamate for more alkaline GABA[76,78]. This figure was created using Biorender.com.

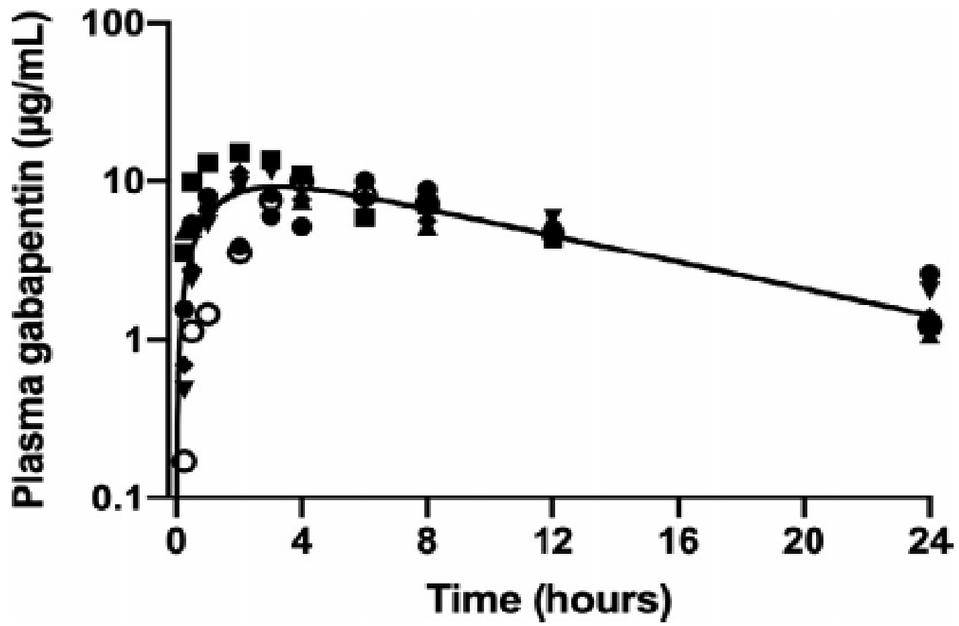
Braga et al., 2024, Gamma-aminobutyric Acid as a potential postbiotic mediator in the gut-brain axis, npi Science of food, 8:16, https://doi.org/10.1038/s41538-024-00253-2

# Pharmacokinetic of GABA

- Liver is the primary metabolic sites for extra-cerebral GABA
- Clinical trial in human, 2 g of GABA 3times daily for 7d
  - Tmax: 0.5-1hr, Half life=5hr = rapid absorption
  - No accumulation of GABA after repeat oral administration
- Commercial sources of GABA:
- 1) microbial fermentation (*L. plantarum* VTTE_133328)
- 2) enzymatic biocatalysis (glutamic acid+enzymatic biocatalysis-Glutamate decarboxylase[GAD] + coenzyme pyridoxal-5´-phosphate)
- 3) chemical synthesis: carboamination reaction of alkenes, catalyzed by copper complex transfer, faster and higher yield but toxic for food and feed

Ahmad et al., 2025, Fermentation, 11, 84, https//:doi.org/10.3390/fermentation11020084
Oketch-Raban et al., 2021, USP safety review of GABA, Nutrients, 13, 2742

# Pharmacokinetic of GABA in Pig




| Parameter | Estimate | Unit | |
|---|---|---|---|
| $t_{1/2}k_a$ | 58 | min | |
| $t_{1/2}$ | 429 | min | 7.15hr |
| $T_{max}$ | 194 | min | 3.23hr |
| $C_{max}$ | 9155 | ng/mL | |
| AUC | 7752303 | min*ng/mL | |
| F Bioavailability | 47 | % | |

**FIGURE 2** Fitted non-linear pharmacokinetic model for gabapentin administered through the oral route to 6 adult Duroc pigs at a dose of 20 mg/kg. The symbols represent the observed plasma gabapentin concentration in each pig, and the line represents the plasma gabapentin concentration predicted by the population model

Blood samples (3 ml) were collected using a 3 ml luer lock disposable syringe (BD Medical, Franklin Lakes, NJ, USA) and a 21G 1.5" hypodermic needle (BD Medical, Franklin Lakes, NJ, USA) from the contralateral caudal auricular vein, or via a 20G 3.5" Quincke spinal needle from the cranial vena cava prior to, and 0.25, 0.5, 1, 2, 3, 4, 6, 8, 12, and 24 h (OS)

Hampton et al., 2021, Pharmacokinetics of oral and compounded intravenous gabapentin in Duroc swine (Sus Scrofa), J Vet Pharmacol Therap, 00:1-7

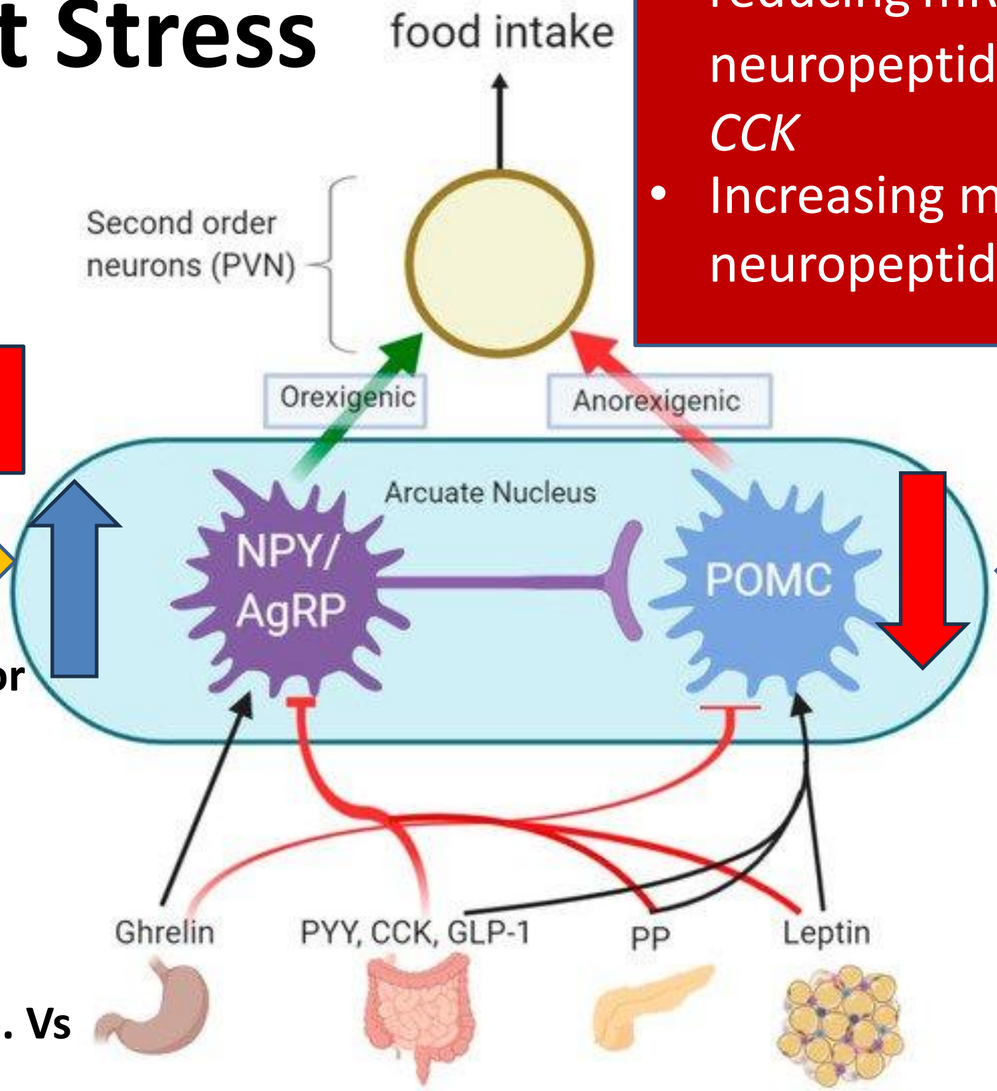
# Poultry-Heat Stress

During HS condition: GABA supplement may

- reducing mRNA expression level of FI-inhibiting neuropeptides including *POMC, leptin, Ghrelin, CCK*
- Increasing mRNA expression of FI-stimulating neuropeptides including *AgRP, NPY*

HS

GABA




GABA

**Broiler received GABA 100ppm for 32ds under both normal and HS conditions Vs CON**

- **Increasing FI**
- **Increasing BW&WG**

**GABA group under HS** (33±2°C) **: reducing the elevated body temp. Vs CON**

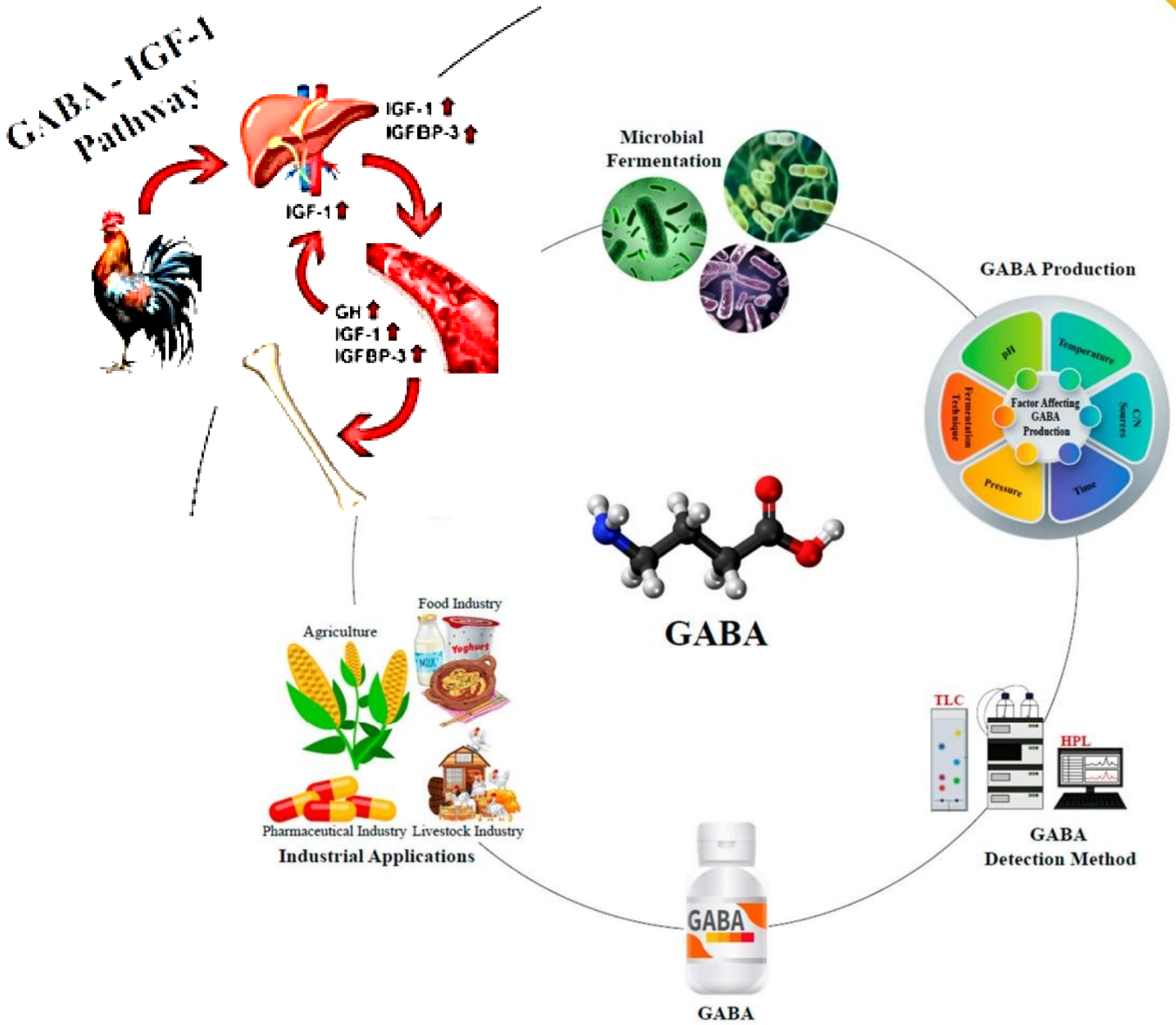
GABA modulate the hypothalamic appetite-regulation peptides

Figure source: nutrients-13-00481-g002

El-Naggar et al., 2019, Role of GABA in regulating feed intake in commercial broilers reared under normal and heat stress conditions, Journal of Thermal Biology, 84:164-175

# GABA-IGF-1 Partway

GABA regulated IGF-1 (Insulin-like-growth factor-1)
IGF-1 is an important peptide hormone for signaling and being growth factor to promote metabolism, cartilage-bone-muscle growth
IGF-1 is primary produced by liver and regulated by growth hormone (GH), dietary condition, and IGF-binding proteins (IGFBP)

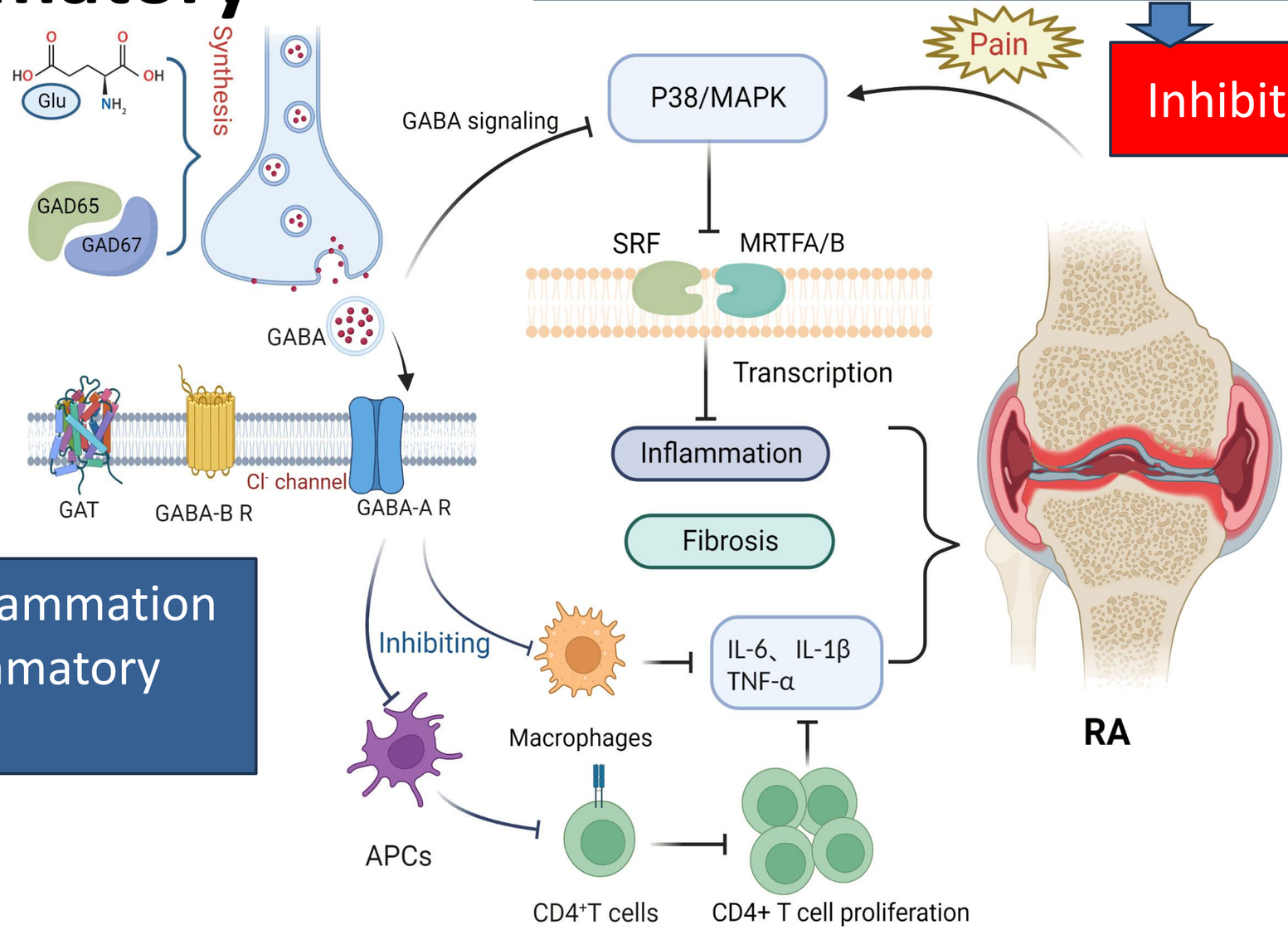



Ahmad et al., 2025, Fermentation, 11, 84,
https//:doi.org/10.3390/fermentation11020084

# Anti-inflammatory

GABA inhibit pain by inhibiting P38/MAPK

Inhibition of pain

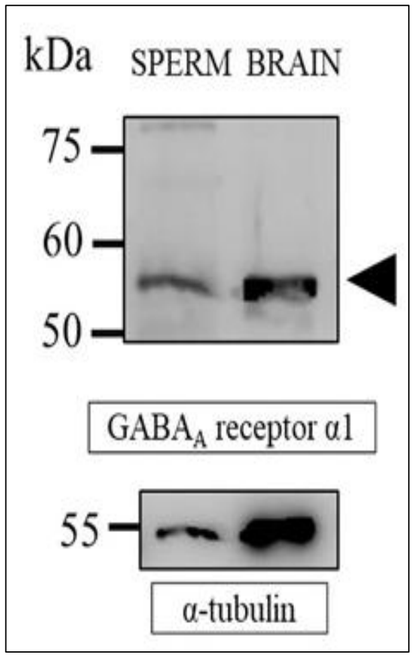



GABA provide anti-inflammation by inhibiting proinflammatory cytokine

Shan et al., 2023, Understanding the function of the GABAergic system and its potential role in rheumatoid arthritis. Front. Immunol. 14:1114350. doi: 10.3389/fimmu.2023.1114350

# GABA and the Reproductive System

- GABA supplement can enhance capacitation of sperms, modulate placenta trophoblasts and stimulate androgen production in human
- GABA may be involved in the fertilization process by supporting the release of enzyme for sperm to penetrate the egg vestment and fuse with the oocyst plasma membrane
- Recent study by Kurata et.al., 2022, the western blot can detect $GABA_A$ receptor alpha1 subunit in porcine sperms




Kurata et al., 2022, Exogenous GABA addition enhance porcine sperm acrosome reaction, Animal Science Journal, 93:e13744, https://doi.org/10.1111/asj.13744
Oketch-Raban et al., 2021, USP safety review of GABA, Nutrients, 13, 2742

# GABA benefit for animals & field data

# Advantage of GABA supplement in poultry

- Reduction of heat stress and high-density stress effect
- Improve growth performance and gut immunity
- Regulating of lipid metabolism (reduction of triglyceride content) and reducing liver fat deposition induced by high fat diet in broilers
- Reducing the dysbiosis of gut microflora of broilers fed with high fat diet
  - Increasing the relative abundance of Bacteriodes phylum and Barnesiella genus
  - Decreasing the relative abundance of Firmicutes phylum and Ruminococcus_torques_groups and Romboutsia genus
  - Promotion of SCFAs such as propionic acid and butyric acid in cecal contents

Chen et al., 2015, The Journal of Animal&Plant Sciences, 25(1):78-87
Chen et al., 2022, Microorganisms, 10, 1281
El-Naggar et al., 2019, Journal of Thermal Biology, 84:164-175
Ka et al., 2014, http://www.journal.msu.ac.th/index.php?page=show_journal_no&no_id=153
Park et al., 2017, J.Sci.Food Agric., 97(9):2891-2897

Chen et al., 2015, Protective effect of GABA on the small intestinal mucosa in heat-stressed Wenchang chicken, The Journal of Animal & Plant Sciences, 25(1):78-87

# Broiler-Heat Stress (HS)

GABA 1mg/bird/dx15ds

GABA improve gut immunity in broiler under HS

Goblet cells: Duo&Jeju HS+GABA > HS&Con

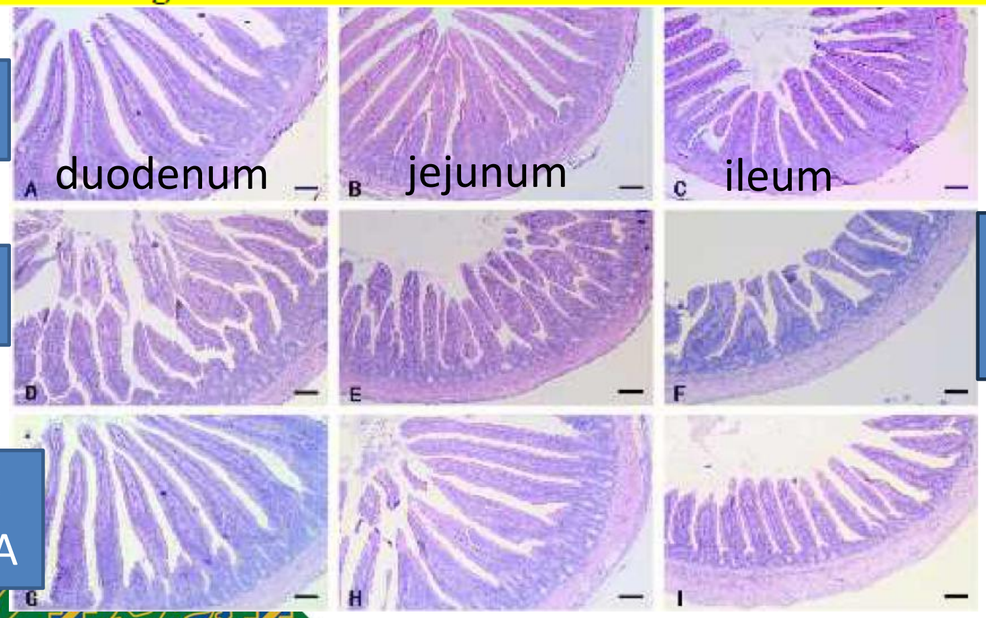
Goblet cells: Ileum HS+GABA > HS

**Table 2. Comparison of the number of intestinal mucosal epithelial goblet cells among the control and experimental groups[1] (cells/100 IEC)**

| Items | Groups | 3d | 6d | 9d | 12d | 15d |
|---|---|---|---|---|---|---|
| Duodenum | Control | 39.33±3.11$^{ab}$ | 37.67±1.67$^{a}$ | 41.67±5.78$^{a}$ | 34.67±5.56 | 39.75±2.88$^{a}$ |
| | HS | 33.67±2.89$^{b}$ | 33.17±1.68$^{b}$ | 29.67±3.78$^{b}$ | 32.03±2.67 | 35.33±1.56$^{b}$ |
| | GABA+HS | 45.33±1.56$^{a}$ | 39.50±1.50$^{a}$ | 43.67±2.89$^{a}$ | 33.33±1.11 | 42.02±4.40$^{a}$ |
| | Sign | NS | * | * | NS | * |
| Jejunum | Control | 47.03±1.33$^{a}$ | 52.60±2.88$^{a}$ | 49.20±1.64$^{a}$ | 39.83±2.50$^{a}$ | 46.05±2.13$^{a}$ |
| | HS | 40.04±1.33$^{b}$ | 42.03±1.33$^{b}$ | 41.80±5.36$^{b}$ | 33.60±4.23$^{b}$ | 37.80±2.24$^{b}$ |
| | GABA+HS | 50.33±1.78$^{a}$ | 54.20±3.50$^{a}$ | 50.25±2.25$^{a}$ | 37.33±1.56$^{a}$ | 49.67±3.11$^{a}$ |
| | Sign | * | * | * | * | * |
| Ileum | Control | 57.50±3.25$^{ab}$ | 60.01±1.50$^{a}$ | 60.80±2.59 | 56.75±3.88 | 62.33±1.78$^{a}$ |
| | HS | 52.40±4.32$^{b}$ | 48.02±2.12$^{b}$ | 58.20±3.84 | 53.02±3.26 | 49.67±4.22$^{b}$ |
| | GABA+HS | 61.25±1.75$^{a}$ | 60.04±1.51$^{a}$ | 63.33±4.89 | 53.80±1.80 | 53.50±2.06$^{ab}$ |
| | Sign | NS | * | NS | NS | NS |

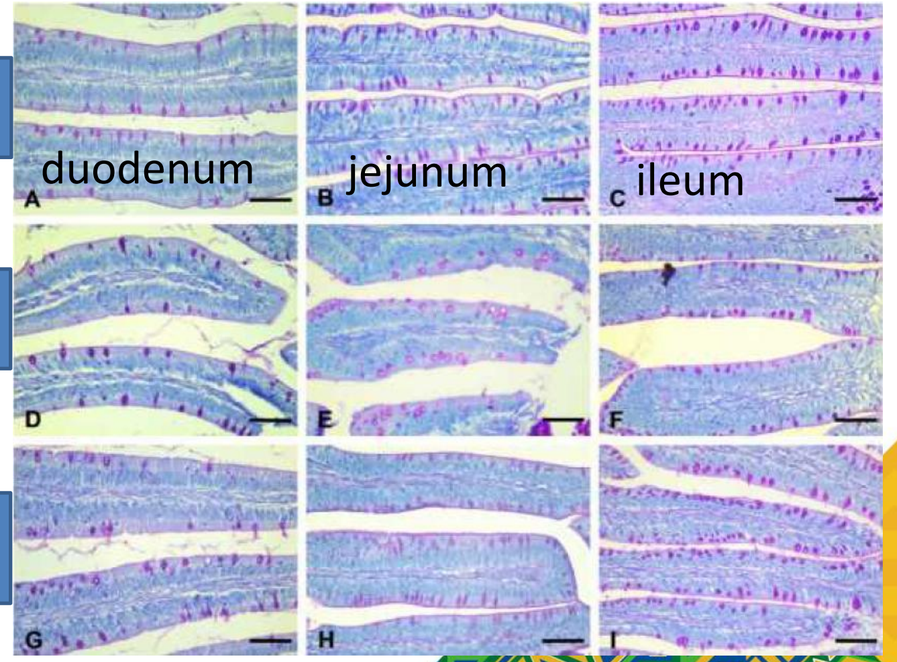
[1] In the same rank, data with different superscript letters indicate statistically significant difference. * ($p<0.05$)

**HE staining of the sections derived from chicken intestine tissues.**




Thin epithelium and shorter villi

Goblet cells




More Goblet cells

# GABA supplement in broilers-HF diet & Fat Metabolism (100ppm GABA*40days)

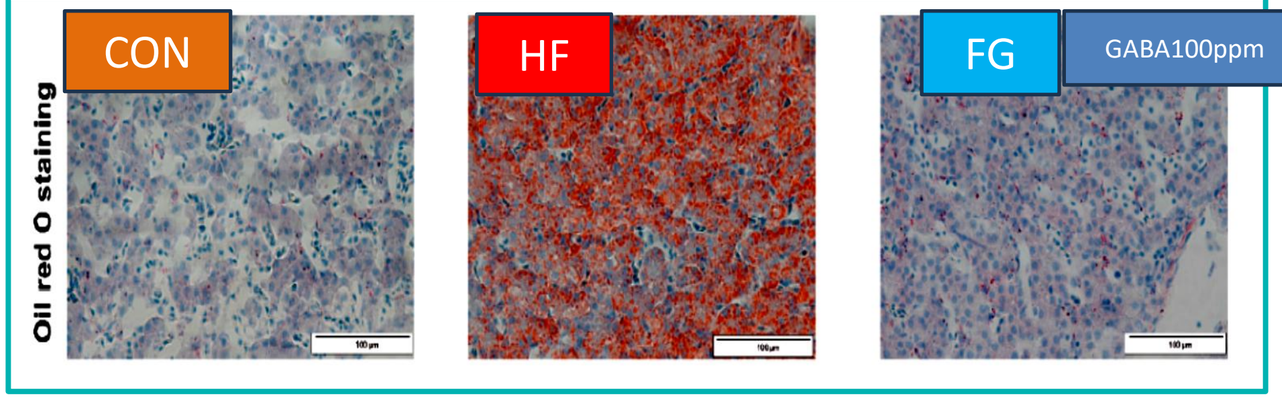
TC=Total cholesterol; TG=Triglyceride

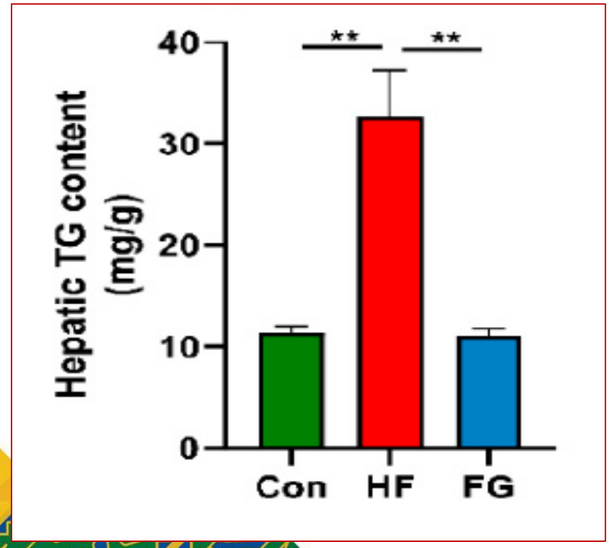
Table 3. Effect of high-fat diet and GABA supplementation on the serum lipid profile in broilers.

| Parameters [1] | Con | HF | FG |
|---|---|---|---|
| HDL-C (mmol/L) | 2.47 ± 0.16 | 2.59 ± 0.16 | 2.69 ± 0.12 |
| LDL-C (mmol/L) | 0.83 ± 0.06 | 0.68 ± 0.04 | 0.68 ± 0.07 |
| TC (mmol/L) | 3.23 ± 0.17 | 3.24 ± 0.19 | 3.24 ± 0.17 |
| TG (mmol/L) | 0.561 ± 0.043 [a] | 0.808 ± 0.119 [b] | 0.544 ± 0.052 [a] |

[1] low-density lipoprotein cholesterol. [a,b] Mean with different superscripts in the same row differ significantly ($p < 0.05$).

Con=control diet; HF=high fat diet; FG=Feeding GABA in high fat diet



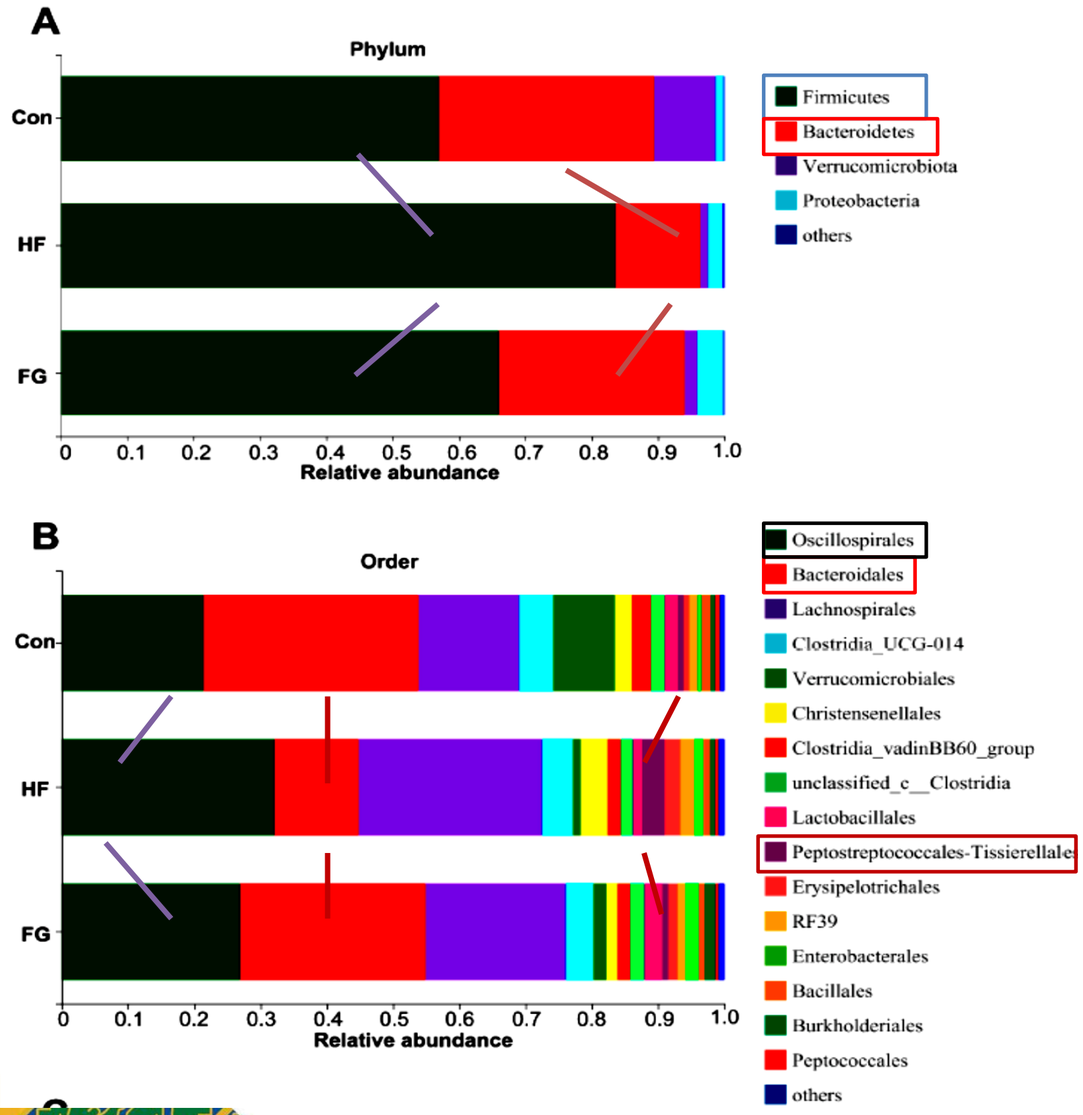



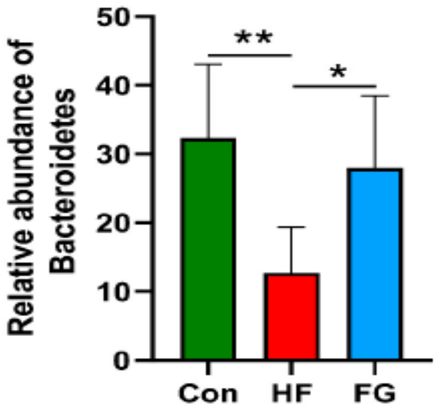

Staining of liver section showed that HF diets induce obvious hepatocyte vacuolation and increased liver lipid droplets accumulation, GABA supplement significantly reduced hepatic steatosis (fatty liver) in broiler fed HF diets

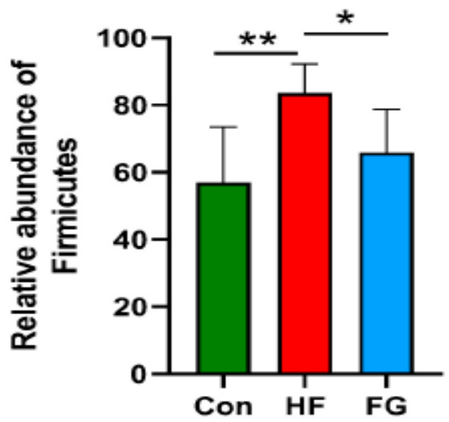
Oil Red O staining is for staining the lipid droplet, it can be used for staining of triglycerides and low polar lipids

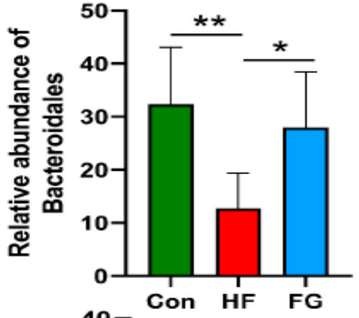
Chen et al., 2022, Dietary GABA supplementation inhibits high-fat-diet-induced hepatic steatosis via modulation gut microbiota in broilers, Microorganisms, 10, 1281

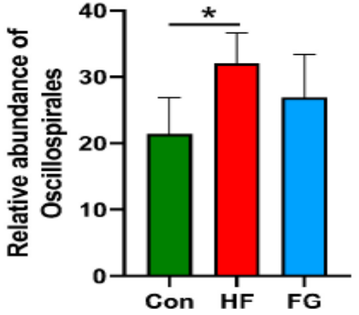
# GABA supplement in broilers-microbiotas

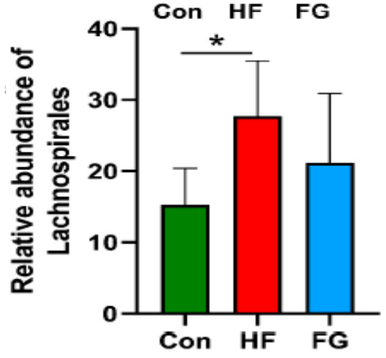


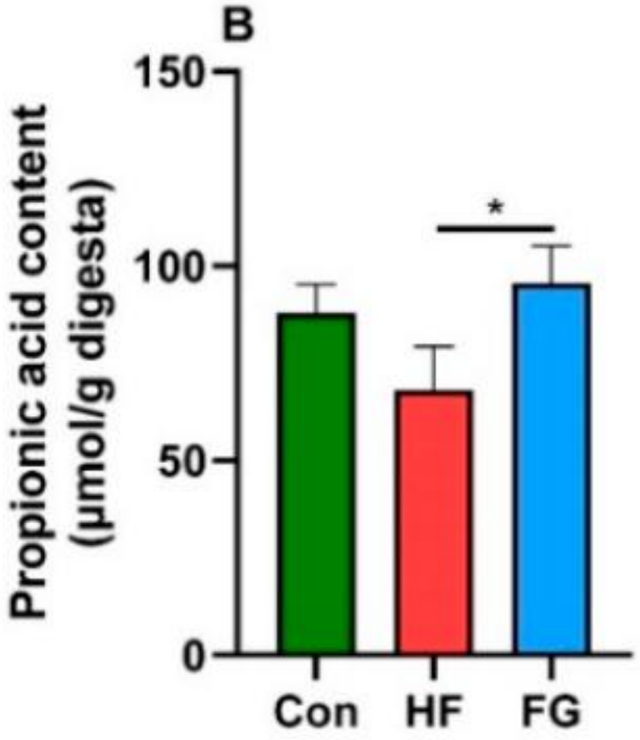



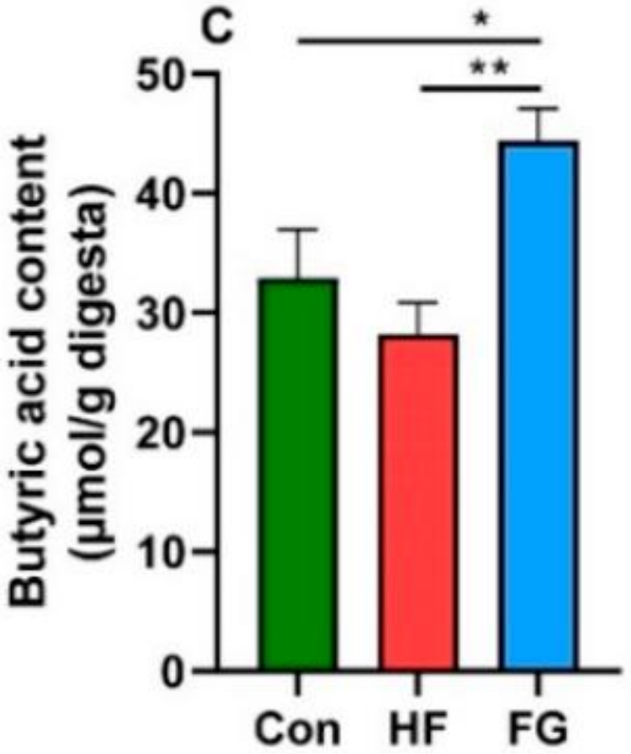

**FG=GABA 100ppm**
***40ds**

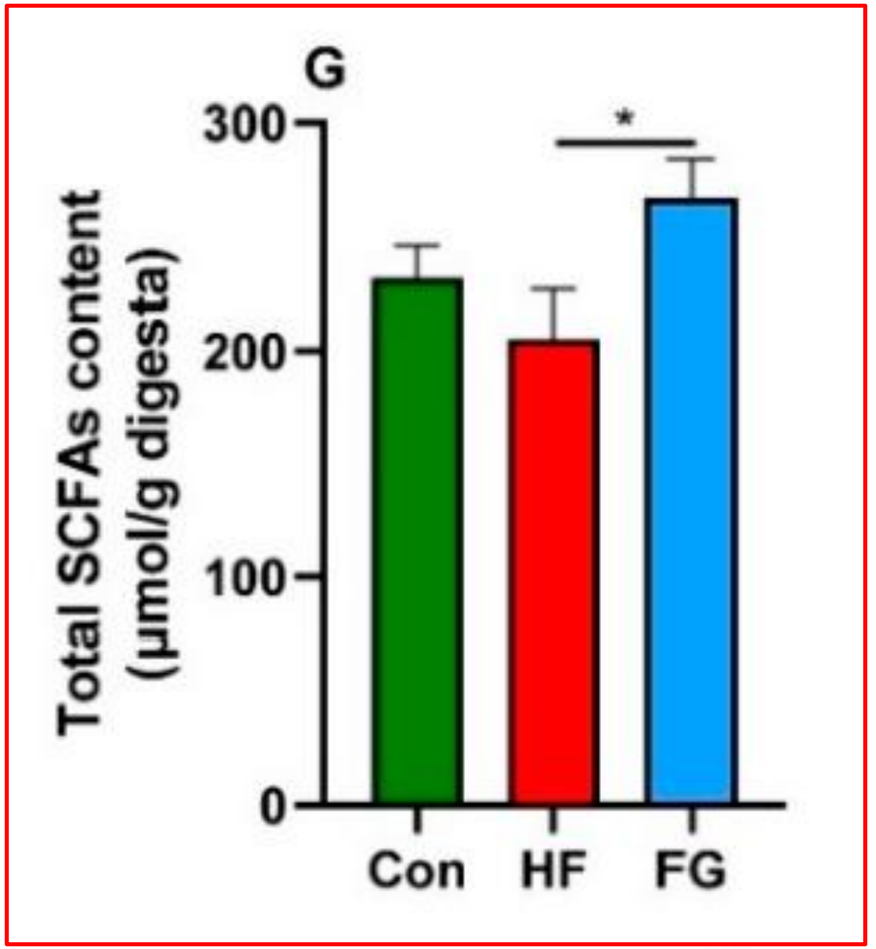
Chen et al., 2022, Dietary GABA supplementation inhibits high-fat-diet-induced hepatic steatosis via modulation gut microbiota in broilers, Microorganisms, 10, 1281

Con=control diet; HF=high fat diet; FG=Feeding GABA in high fat diet

# GABA supplement in broilers-SCFAs




FG=GABA 100ppm *40ds

Chen et al., 2022, Dietary GABA supplementation inhibits high-fat-diet-induced hepatic steatosis via modulation gut microbiota in broilers, Microorganisms, 10, 1281

Con=control diet; HF=high fat diet; FG=Feeding GABA in high fat diet

# Layer-Productivity & Immunity

Table 2. The effect of dietary gamma-aminobutyric acid (GABA) on productivity in layers[1]

| | GABA (ppm) | | | | SEM[2] | *P*-value | |
|---|---|---|---|---|---|---|---|
| | 0 | 25 | 50 | 100 | | linear | quadratic |
| 32–36 week | | | | | | | |
| Egg production (%) | 93.5 | 95.2 | 95.4 | 97.1 | 0.48 | < 0.001 | 0.931 |
| Egg weight (g) | 62.2 | 64.3 | 65.9 | 66.8 | 0.15 | < 0.001 | 0.348 |
| Egg mass (g/day/hen) | 58.2 | 61.2 | 62.9 | 64.9 | 0.35 | < 0.001 | 0.643 |
| Feed intake (g) | 127 | 125 | 131 | 130 | 0.79 | 0.536 | 0.943 |
| Feed conversion (g:g) | 2.18 | 2.04 | 2.08 | 2.00 | 0.02 | 0.279 | 0.558 |

32 wks old layers, starting weight is 1,946±0.5g GABA supplement in feed at 0-100ppm for 5wk

GABA increase layer performance

GABA monitoring humeral immunity such as haptoglobin and IgG

Haptoglobin is the acute phase protein increasing after infection

Table 4. The effect of dietary gamma-aminobutyric acid (GABA) on blood profiles in layers[1]

| | GABA (ppm) | | | | SEM[2] | *P*-value | |
|---|---|---|---|---|---|---|---|
| | 0 | 25 | 50 | 100 | | linear | quadratic |
| WBC (K/μl) | 311 | 318 | 324 | 329 | 14.3 | 0.656 | 0.978 |
| RBC (M/μl) | 2.41 | 2.24 | 2.37 | 2.33 | 0.04 | 0.786 | 0.512 |
| Lymphocyte (%) | 54.3 | 55.1 | 57.0 | 58.5 | 1.94 | 0.438 | 0.938 |
| Haptoglobin (μg/ml) | 150 | 123 | 110 | 108 | 5.52 | 0.014 | 0.289 |
| IgG (μg/ml) | 488 | 540 | 543 | 683 | 36.64 | 0.044 | 0.545 |
| Cortisol (ng/ml) | 2.74 | 2.54 | 2.62 | 2.56 | 0.15 | 0.385 | 0.814 |
| Epinephrine (pg/ml) | 91.5 | 87.9 | 82.1 | 74.2 | 4.82 | 0.280 | 0.854 |
| Norepinephrine (pg/ml) | 179.3 | 153.2 | 143.5 | 141.8 | 9.85 | 0.204 | 0.563 |

Park&Kim, 2015, Effect of dietary GABA on egg production, egg quality, and blood profiles in layer hens, Veterinarni Medicina, 6 (11):629-634

# GABA supplement via Germinated Paddy Rice (GPR) effect in layer hens

**Table 3.** Effect of dietary GPR levels on egg performance and egg quality in laying hens during 42 to 56 week of ages.

| Item | | | ADFI, g/b | AEW, g | HD, % | EM, g/b/d | SS, gf/m$^2$ | GABA in Egg, mg/100 g | GABA supplement in feed mg/kg |
|---|---|---|---|---|---|---|---|---|---|
| Treatment | Stocking density | GPR, g/kg | | | | | | | |
| T1 | LSD | 0 | 96.9 | 46.7 | 80.5 | 37.6 | 2657.4 $^{b}$ | 9.5 | 0 |
| T2 | | 74 | 93.3 | 51.7 | 82.2 | 42.5 | 2732.6 $^{b}$ | 10.3 | 3.1 |
| T3 | | 148 | 91.5 | 50.3 | 82.0 | 41.2 | 3454.6 $^{a}$ | 12.8 | 6.3 |
| T4 | HSD | 0 | 83.7 | 47.0 | 76.6 | 36.0 | 2062.0 $^{c}$ | 8.4 | 0 |
| T5 | | 74 | 82.6 | 48.0 | 76.0 | 36.5 | 2553.6 $^{b}$ | 8.9 | 3.1 |
| T6 | | 148 | 82.0 | 49.5 | 73.6 | 36.4 | 2795.1 $^{b}$ | 12.1 | 6.3 |

Hy-line Brown laying hen fed 18% CP, 2,800 kcal/kg diet
From 42-56WOA: feed with GPR at 0, 74 and 148 g/kg contained GABA 0, 3.1, and 6.3 mg/kg diet in LSD & HSD condition
Daily cyclic temperature : 27-35°C for 6hr; 35±1 °C for 12hr; 35-27 °C for 6hr, RH=65-70%

LSD=1.12m$^2$/bird (8.3 birds/m$^2$)
HSD=0.06m$^2$/bird (16.7 birds/m$^2$)

Supplement GABA in feed, GABA in egg can increase, dose dependently

GPR: germinated paddy rice; LSD: low stocking density; HSD: high stocking density; T1: LSD + 0 g/kg GPR; T2: LSD + 74 g/kg GPR; T3: LSD + 148 g/kg GPR; T4: HSD + 0 g/kg GPR; T5: HSD + 74 g/kg GPR; T6: HSD + 148 g/kg GPR; ADFI: average daily feed intake; AEW: average egg weight; HD: hen-day egg production; EM: egg mass; SS: shell breaking strength; SEM: standard error of the means. $^{a-c}$ Means with different superscripts in a column differ significantly ($p < 0.05$).

Incharoen, et.al., 2021, Dietary germinated paddy rice (GPR) and stocking density affect egg performance, serum biochemical properties and proteomic and transcriptomic response of laying hens exposed to chronic heat stress, Proteomes, 9, 48, https://doi.org/10.3390/proteomes9040048

# GABA WS supplement in laying hens-1

Lohmann Brown laying hens 32 WOA
Water supplement with GABA for 4wks
G1 Control without GABA
G2 GABA40: GABA 40mg/kg bw
G3 GABA80: GABA 80mg/kg bw

Dose Dependence

**Table (2):**- Effect of GABA on laying performance in laying hens during the experimental period.
Values are means ± SE. Means within the same row of different letters are significantly different at ($P \leq 0.05$).

| Parameters | Experimental groups | | | |
|---|---|---|---|---|
| | Weeks | Group (1) | Group (2) | Group (3) |
| Egg Production (%) | 1st wk | 57.00±1.06$^{a}$ | 67.85±3.71$^{a}$ | 77.28±3.09$^{a}$ |
| | 2nd wk | 55.85±0.40$^{ab}$ | 75.28±3.85$^{b}$ | 84.14±1.33$^{a}$ |
| | 3rd wk | 59.42±0.71$^{ab}$ | 78.42±2.21$^{b}$ | 89.85±1.84$^{a}$ |
| | 4th wk | 63.00±1.11$^{c}$ | 80.14±2.42$^{b}$ | 92.01±1.46$^{a}$ |
| Egg weight (g) | 1st wk | 54.42±0.57$^{a}$ | 57.00±1.06$^{a}$ | 58.14±1.99$^{a}$ |
| | 2nd wk | 55.14±0.50$^{b}$ | 55.85±0.40$^{ab}$ | 57.14±0.50$^{a}$ |
| | 3rd wk | 57.28±0.42$^{b}$ | 59.42±0.71$^{ab}$ | 62.28±1.71$^{a}$ |
| | 4th wk | 59.28±0.47$^{b}$ | 63.00±1.11$^{ab}$ | 65.14±2.97$^{a}$ |
| Egg mass (g/ hen/ d) | 1st wk | 36.84±1.80$^{b}$ | 41.10±0.42$^{a}$ | 42.62±0.48$^{a}$ |
| | 2nd wk | 41.56±2.33$^{b}$ | 47.07±1.35$^{a}$ | 50.97±0.79$^{a}$ |
| | 3rd wk | 45.04±1.59$^{b}$ | 48.08±2.88$^{ab}$ | 53.73±1.16$^{a}$ |
| | 4th wk | 49.80±1.65$^{b}$ | 51.98±0.68$^{b}$ | 58.84±0.79$^{a}$ |
| FCR | 1st wk | 2.10±0.11$^{a}$ | 2.02±0.29$^{a}$ | 1.90±0.18$^{a}$ |
| | 2nd wk | 2.44±0.13$^{a}$ | 2.07±0.06$^{a}$ | 2.04±0.18$^{a}$ |
| | 3rd wk | 2.40±0.07$^{a}$ | 2.16±0.07$^{b}$ | 1.90±0.04$^{c}$ |
| | 4th wk | 2.31±0.08$^{a}$ | 2.12±0.07$^{ab}$ | 2.00±0.08$^{b}$ |

**Table (3):** Effect of GABA supplementation on external egg quality parameters in laying hens during the experimental period.
Values are means ± SE. Means within the same row of different letters are significantly different at ($P \leq 0.05$).

| Parameters | Experimental groups | | | |
|---|---|---|---|---|
| | Weeks | Group (1) | Group (2) | Group (3) |
| Egg shape index (%) | 1st wk | 0.76±0.007$^{a}$ | 0.77±0.004$^{a}$ | 0.78±0.016$^{a}$ |
| | 2nd wk | 0.75±0.010$^{b}$ | 0.77±0.007$^{a}$ | 0.79±0.015$^{a}$ |
| | 3rd wk | 0.75±0.010$^{b}$ | 0.77±0.008$^{b}$ | 0.78±0.005$^{a}$ |
| | 4th wk | 0.76±0.021$^{b}$ | 0.78±0.005$^{ab}$ | 0.80±0.004$^{a}$ |
| Shell weight (gm) | 1st wk | 6.89±0.35$^{a}$ | 7.11±0.10$^{a}$ | 6.93±0.017$^{a}$ |
| | 2nd wk | 7.12±0.19$^{b}$ | 7.25±0.14$^{b}$ | 7.80±0.16$^{a}$ |
| | 3rd wk | 8.01±0.13$^{b}$ | 8.19±0.20$^{b}$ | 8.65±0.09$^{a}$ |
| | 4th wk | 7.81±0.29$^{b}$ | 7.99±0.14$^{b}$ | 8.76±0.08$^{a}$ |
| Shell thickness (mm) | 1st wk | 0.42±0.011$^{a}$ | 0.43±0.009$^{a}$ | 0.45±0.019$^{a}$ |
| | 2nd wk | 0.40±0.003$^{b}$ | 0.43±0.017$^{ab}$ | 0.45±0.009$^{a}$ |
| | 3rd wk | 0.40±0.006$^{b}$ | 0.44±0.014$^{a}$ | 0.45±0.010$^{a}$ |
| | 4th wk | 0.46±0.011$^{b}$ | 0.48±0.013$^{ab}$ | 0.51±0.020$^{a}$ |

More egg production % from Wk2 in both conc.
Higher egg weight from wk3 in GABA80
Higher egg mass from wk1 in both conc.
Better FCR from wk3 in both conc.
Higher shell thickness from wk3 in both conc.

Elkomy et al., 2020, Effect of drinking water supplemented with GABA on laying performance and egg quality parameters in laying hens, Int.J.Adv.Res, 8(02):312-318

# GABA WS supplement in laying hens-2

**Table (4):**- Effect of GABA supplementation on egg yolk quality in laying hens during the experimental period.

| Parameters | Experimental groups | | | |
|---|---|---|---|---|
| | Weeks | Group (1) | Group (2) | Group (3) |
| Yolk weight (gm) | 1st wk | 18.78±1.43$^{a}$ | 20.07±0.60$^{a}$ | 19.84±0.45$^{a}$ |
| | 2nd wk | 19.27±0.95$^{b}$ | 20.37±1.22$^{b}$ | 26.14±1.44$^{a}$ |
| | 3rd wk | 21.33±0.43$^{b}$ | 22.45±0.10$^{ab}$ | 23.89±0.84$^{a}$ |
| | 4th wk | 22.21±0.57$^{b}$ | 24.01±0.44$^{a}$ | 26.36±0.66$^{a}$ |
| Yolk index% | 1st wk | 33.93±2.98$^{a}$ | 35.11±1.60$^{a}$ | 38.01±0.76$^{a}$ |
| | 2nd wk | 35.91±2.13$^{b}$ | 39.80±2.86$^{ab}$ | 45.33±1.84$^{a}$ |
| | 3rd wk | 39.66±4.59$^{b}$ | 41.42±3.89$^{ab}$ | 51.18±1.37$^{a}$ |
| | 4th wk | 40.30±3.25$^{b}$ | 44.38±3.34$^{ab}$ | 49.29±0.82$^{a}$ |
| Yolk color | 1st wk | 9.42±0.42$^{a}$ | 9.85±0.34$^{a}$ | 10.28±0.81$^{a}$ |
| | 2nd wk | 9.00±0.48$^{b}$ | 9.57±0.52$^{b}$ | 11.00±0.37$^{a}$ |
| | 3rd wk | 9.85±0.34$^{b}$ | 10.14±0.34$^{ab}$ | 11.14±0.35$^{a}$ |
| | 4th wk | 8.42±0.20$^{b}$ | 9.14±0.51$^{ab}$ | 9.71±0.42$^{a}$ |
| Yolk cholesterol (mg/whole yolk) | 1st wk | 249.41±19.22$^{a}$ | 205.57±10.60$^{b}$ | 168.21±4.28$^{b}$ |
| | 2nd wk | 247.69±9.61$^{a}$ | 206.31±5.30$^{b}$ | 164.05±2.10$^{c}$ |
| | 3rd wk | 140.28±3.74$^{a}$ | 123.44±5.41$^{b}$ | 115.48±0.68$^{b}$ |
| | 4th wk | 138.85±2.35$^{a}$ | 120.58±2.71$^{b}$ | 114.05±1.57$^{b}$ |
| Triglycerides (mg/whole yolk) | 1st wk | 233.85±15.03$^{a}$ | 191.68±9.17$^{b}$ | 153.00±4.19$^{c}$ |
| | 2nd wk | 230.99±6.45$^{a}$ | 190.25±5.58$^{b}$ | 151.57±2.18$^{c}$ |
| | 3rd wk | 127.43±3.75$^{a}$ | 109±4.57$^{b}$ | 105.84±1.46$^{b}$ |
| | 4th wk | 125.36±2.32$^{a}$ | 110.36±1.97$^{b}$ | 106.55±0.71$^{b}$ |

Values are means ± SE. Means within the same row of different letters are significantly different at ($P \leq 0.05$).

G1 Control without GABA
G2 GABA40: GABA 40mg/kg bw
G3 GABA80: GABA 80mg/kg bw

Dose Dependence

**Table (5):** Effect of GABA on albumen quality of laying hens during the experimental period.

| Parameters | Experimental groups | | | |
|---|---|---|---|---|
| | Weeks | Group (1) | Group (2) | Group (3) |
| Albumen weight (gm) | 1st wk | 29.59±2.22$^{a}$ | 30.86±0.94$^{a}$ | 32.65±1.36$^{a}$ |
| | 2nd wk | 36.79±1.58$^{b}$ | 38.66±1.05$^{ab}$ | 41.50±1.31$^{a}$ |
| | 3rd wk | 38.24±0.67$^{b}$ | 41.92±2.26$^{ab}$ | 44.14±1.93$^{a}$ |
| | 4th wk | 38.43±1.63$^{b}$ | 38.78±1.39$^{b}$ | 43.82±1.05$^{a}$ |
| Albumen % | 1st wk | 52.88±3.12$^{a}$ | 54.28±0.52$^{a}$ | 58.03±1.11$^{a}$ |
| | 2nd wk | 54.04±3.25$^{b}$ | 55.76±3.10$^{b}$ | 64.25±1.62$^{a}$ |
| | 3rd wk | 60.87±0.61$^{b}$ | 61.51±1.28$^{b}$ | 66.95±1.29$^{a}$ |
| | 4th wk | 61.11±2.12$^{b}$ | 62.36±2.73$^{b}$ | 70.65±1.73$^{a}$ |
| Albumen haugh unit | 1st wk | 106.12±1.09$^{b}$ | 107.86±1.31$^{ab}$ | 115.15±1.52$^{a}$ |
| | 2nd wk | 109.67±0.97$^{b}$ | 110.75±2.07$^{b}$ | 117.35±1.67$^{a}$ |
| | 3rd wk | 110.99±1.63$^{b}$ | 113.34±2.69$^{b}$ | 120.58±2.15$^{a}$ |
| | 4th wk | 113.88±2.33$^{b}$ | 115.51±3.74$^{b}$ | 124.62±2.70$^{a}$ |

Values are means ± SE. Means within the same row of different letters are significantly different at ($P \leq 0.05$).

Better yolk & albumin quality
Lower yolk cholesterol and TG from wk1 in both conc.

Elkomy et al., 2020, Effect of drinking water supplemented with GABA on laying performance and egg quality parameters in laying hens, Int.J.Adv.Res, 8(02):312-318

# GABA in Duck (Cherry Valley)

**Table 3: Effect of dietary γ-aminobutyric acid (GABA) on feed intake (FI, n=5) survival rate (n=5), body weight (n=5), feed conversion ratio (FCR, n=5) and dressing percentage (n=20) in heat stressed ducks at 42 days**

| Parameters | Control | GABA | P-value |
|---|---|---|---|
| FI (20-32 days, g/bird/d) | 167.1 ± 3.84[a] | 147.91 ± 2.56[b] | 0.0032 |
| FI (32-42 days, g/bird/d) | 189.49 ± 1.66[a] | 158.22 ± 4.75[b] | 0.0003 |
| Survival rate (%) | 96.78 ± 0.34 | 97.00 ± 0.24 | 0.6085 |
| Initial body weight (kg) | 0.95 ± 0.02 | 0.98 ± 0.04 | 0.0824 |
| Final body weight (kg) | 2.60 ± 0.01 | 2.65 ± 0.02 | 0.0824 |
| FCR (kg/kg) | 2.15 ± 0.04[a] | 1.96 ± 0.02[b] | 0.0042 |
| Dressing percentage (%) | 71.60 ± 0.02 | 71.63 ± 0.03 | 0.3721 |

[a, b] Means with different letters in a row differ significantly (P<0.05)

**Table 5: Effect of dietary γ-aminobutyric acid (GABA) on GABA concentrations in blood and different tissues of heat stressed ducks (n=20) at 42 days**

| Parameters | Control | GABA | P-value |
|---|---|---|---|
| Blood (nmol/ml) | 13.17 ± 2.88 | 15.72 ± 2.22 | 0.4981 |
| Heart (µmol/g) | 0.27 ± 0.02 | 0.32 ± 0.03 | 0.1657 |
| Liver (µmol/g) | 0.59 ± 0.03 | 0.64 ± 0.04 | 0.3365 |
| Kidney (nmol/g) | 18.80 ± 1.21 | 20.23 ± 1.75 | 0.5163 |
| Brain (µmol/g) | 2.91 ± 0.13[b] | 3.51 ± 0.05[a] | 0.0030 |

[a, b] Means with different letters differ significantly (P<0.05)

2,000 Cherry Valley ducks 19do

- 1,000 ducks for control
- 1,000 ducks for GABA supplement group at 100g/ton feed until 42 do

Less feed intake

Less FCR

Higher GABA in organs, especially brain at 42 do

Zhigang etal., 2013, Effect of dietary GABA on performance parameters and some plasma metabolites in Cherry Valley ducks under high ambient temperature, Iranian Journal of Veterinary Research, Shiraz University, 14(4):283-290

# Field Study: Evaluation of GABA Feed Supplement in Broiler

| Growth Performance | GABA | Control | *P*-value | Diff/bird |
|---|---|---|---|---|
| Weight In (g) | 48.5±5.07 | 47.33±5.05 | 0.730 | 1.17g |
| Weight out (g) | 2,800±0 | 2,633.3±40.8 | <0.001 | 166.7g |
| Days to slaughter | 43.5±1.73 | 43.67±0.52 | 0.826 | -0.17d |
| Weight Gain (g) | 2,751.5±5.07 | 2,586.0±42.65 | <0.001 | 165.5g |
| ADG (g/d) | 63.38±2.44 | 59.23±1.43 | 0.010 | 4.15g/d |
| Feed Intake (g/d) | 112.31±3.0 | 106.3±7.76 | 0.184 | 6.01g/d |
| FCR | 1.76±0.03 | 1.80±0.14 | 0.773 | -0.04 |

Wk5: the ADG (g/bird/d) of the GABA 100ppm supplement group was significantly higher than control group (*P*<0.05)
and keeping momentum of growth until wk6 [one week after stopping supplement]

Growth promoter effect of GABA

| Gr-ADG | Wk1 | Wk2 | Wk3 | Wk4 | Wk5 | Wk6 |
|---|---|---|---|---|---|---|
| GABA100ppm | 20.5±1.08 | 45.97±0.7 | 72.14±1.90 | 73.39±4.66 | 98.21±2.16 | 70.0±12.45 |
| Control | 20.14±1.51 | 44.67±0.98 | 72.69±2.21 | 75.03±3.20 | 83.10±8.73 | 59.96±8.62 |
| *P*-value | 0.694 | 0.052 | 0.696 | 0.525 | 0.010 | 0.167 |
| Difference (g) | 0.36 | 1.30 | -0.55 | -1.64 | 15.11 | 10.04 |

Makhanon, et.al., 2024, The Proc. of the 21st International Society for Animal Hygiene Conference, 18-20 September 2024, Chiang Mai, Thailand, Innovation Technology for Animal Health and Food Production, p76.

# Advantage of GABA supplement in pig

- Decreasing the effect of stress from heat and transportation
- Improve mucosal immunity in normal and infected condition
- Maintain the performance and carcass quality during stress condition
- Balance the gut microbiotas and increase the metabolism of gut microbiotas in normal situation
- Improve sow reproductive performance and piglet growth

Bi et al., 2020, J. Anim. Physiol.Anim.Nutr.(Berl.), 104(2):590-596
Chen et al., 2020, Food&Function, 11:9067-9074
Zhao et al., 2020, Hindawi Mediators of Inflammation, V.2020, Article ID 7368483

# GABA supplement to weaning pigs

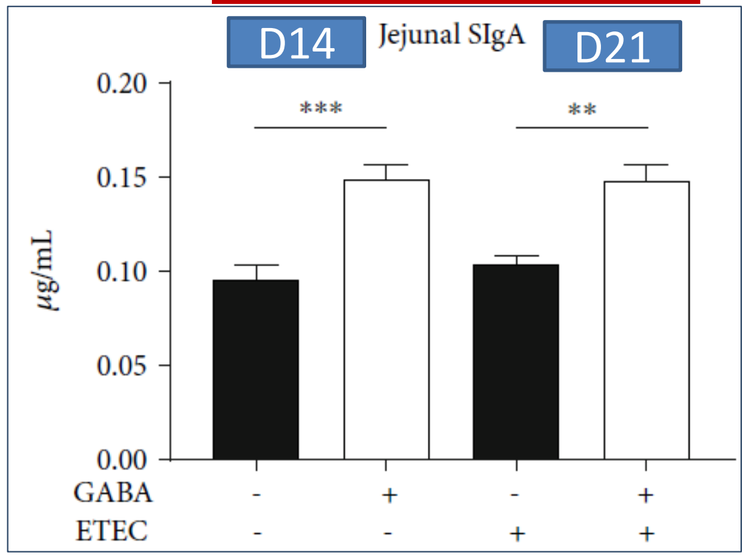
- Study in 21 do 37 piglets at the average body weight 5.82±0.86 kg
- GABA 40 mg/kgx3wk in D1-D21

In normal and ETEC infection

ETEC=Enterotoxigenic *Escherichia coli*

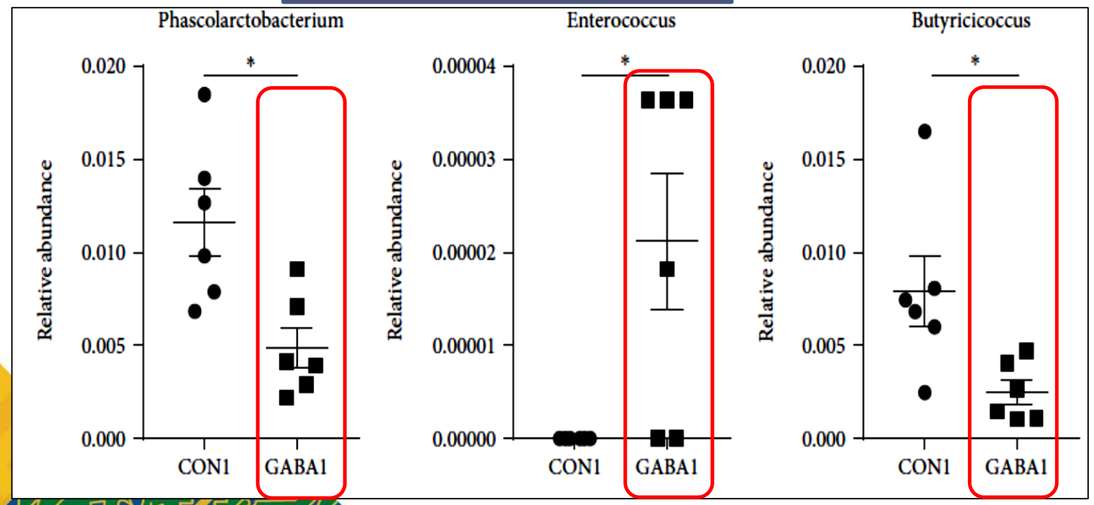
- ELISA also shows GABA supplement increase sIgA in Jejunum for both healthy & ETEC infected pigs
- GABA show significantly role to gut microbiomes in both normal and infected situation

sIgA in intestine

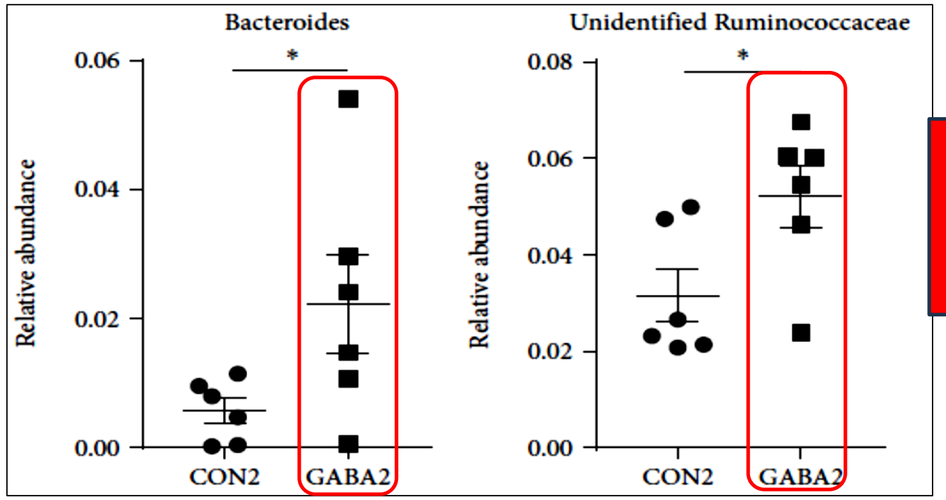



$^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$.

healthy




+ETEC




Gut microbiomes

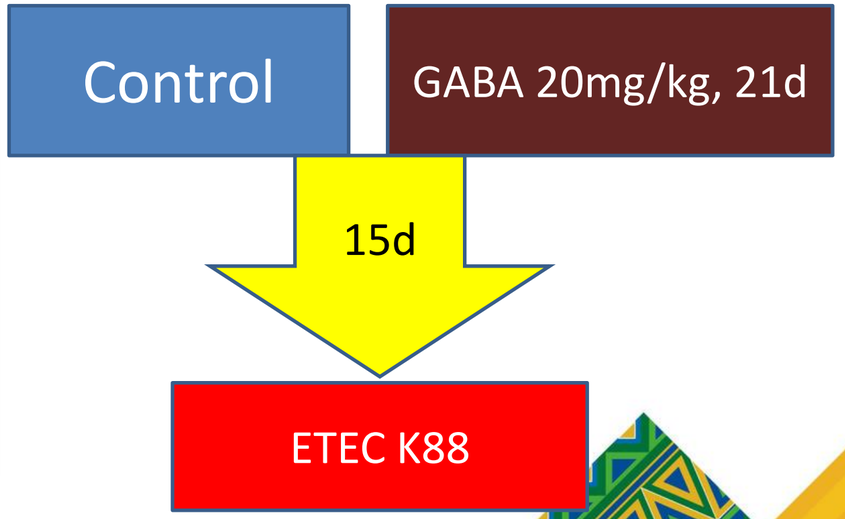
Zhao et al., 2020, Hindawi Mediators of Inflammation, V.2020, Article ID 7368483

# GABA feed supplement to increase growth by moderating serum amino acid and protecting the GIT and microbiomes during ETEC infection

Chen et al. (2020) reported their study of serum amino profile after supplement GABA in piglet feed and the intestinal immune status and microbiota after ETEC K88 challenged

- 18 piglets were divided into 2 groups as control and GABA supplement groups, GABA 20mg/kg feed supplement for 3 weeks
- Day 15 of study, pigs were challenged with ETEC K88

ETEC=Enterotoxigenic *Escherichia coli*




Result: comparing between GABA supplement and control groups, the GABA group showed:

- Reduction of FCR
- Increasing amino acid in serum and ileal mucosa
- Increasing gene expressions of TNF-α, IFN-ɣ, pIgR, MUC2 in ileum
- Changing the ratio of intestinal microbiota

Chen et al., 2020, Food&Function, 11:9067-9074

# GABA and Transportation Stress in Fattening

Performance → ADG higher than control $P<0.01$

FCR less than control $P<0.05$

Stress →

Antioxidant

GSH-Px (Glutathione peroxidase)
GABA > Control, $P<0.05$

Fat oxidation

MDA (Malondialdehyde)
30/T < 0/T, $P<0.05$

GABA 0 and 30 mg/kg feed for 74 days in fattening pigs starting the study at average weight 32.67±0.62 kg

Stress hormone: GABA<Control

Cortisol
30/T < 0/T, $P<0.05$

Adrenal Cortisol Hormone
30/T < 0/T, $P<0.05$

Carcass Quality →

pH45min: 0/T < 30/T , $P<0.05$

Drip Loss: O/T > 30/T, $P<0.05$

Low pH caused protein denature of muscle and water loss from cells and pale muscular color

Bi et al., 2020, J Anim. Physiol. Anim. Nutr. (Berl.), 104 (2):590-596

# GABA supplement in gestating sows

Table 3. Effects of GABA supplementation during late gestation on litter characteristics for the sows.

| Items | CON | GABA | *p* Value |
|---|---|---|---|
| Total piglets (*n*) | 12.50 ± 1.30 | 12.63 ± 0.59 | 0.93 |
| Live piglets (*n*) | 11.75 ± 1.29 | 12.13 ± 0.44 | 0.78 |
| Stillborn (*n*) | 0.75 ± 0.31 | 0.50 ± 0.26 | 0.55 |
| Total litter weight (live, kg) | 16.06 ± 1.83 | 18.79 ± 1.06 | 0.22 |
| Average birth weight (live, kg) | 1.38 ± 0.07 | 1.57 ± 0.10 | 0.14 |
| IUGR piglets [a] (*n*) | 1.00 ±0.38 | 0.25 ±0.16 | 0.09 |

CON = control group fed basal diet, GABA = GABA group fed basal diet plus GABA. *n* = 9 per treatment. [a] Intrauterine growth retardation (IUGR) piglets = piglets with birth weight two standard deviations below the mean birth weight of the population [20]. Values are means ± SEM and analyzed by Student's *t*-test.

Result of study show the improve in reproductive performance, however, it is not statistically significant

Liu et al., 2022, Maternal dietary supplementation with GABA alleviated oxidative stress in gestation sows and their offspring by regulating GABRP, Animals, 12, 2539.

# GAMMA AMINO BUTYRIC ACID SUPPLEMENT IN SOW FEED FOR IMPROVING BREEDER PERFORMANCE

## Reproductive performance from GABA

### Sow reproductive performance between control and GABA feed supplement groups at 100ppm

| Reproductive performance | Control | CV% | GABA supplement | CV% | P-value* |
|---|---|---|---|---|---|
| Wean to service days (WTS) | 10.42±17.14 | 164.49 | 4.42±0.70 | 15.89 | 0.081 |

71.4% of sows received GABA supplement for 60 days of gestation feed + whole lactation have WTS < 5days

### Relation between the GABA supplement days and WTS of sow (*P*-value=0.673)

| 100ppm GABA supplement in sows | N | WTS [N(%)] | | |
|---|---|---|---|---|
| | | Less than 5 days | 5-7 days | No heat |
| Control 0ppm GABA supplement | 26 | 10 (38.5%) | 13 (50%) | 3 (11.5%) |
| GABA 30 days at gestation and whole lactation | 6 | 2 (33.3%) | 4 (66.7%) | 0 |
| GABA 60 days at gestation and whole lactation | 7 | 5 (71.4%) | 2 (28.6%) | 0 |
| GABA 90 days at gestation and whole lactation | 7 | 4 (57.1%) | 3 (42.9%) | 0 |
| GABA >90 days at gestation and whole lactation | 6 | 3 (50%) | 3 (50%) | 0 |
| Total GABA supplement | 26 | 14 (53.8%) | 12 (46.2%) | 0 |

Makhanon et al., 2025, APVS 2025, Program No. P2-027-C

# GABA SUPPLEMENT IN GESTATING AND LACTATING SOW FEED FOR IMPROVING SOW AND PIGLET PERFORMANCE

## Reproductive performance from GABA

**Table 1 Sow farrowing performance**

$^{a,b}$ $P<0.05$

| Details | Group (Total No=300sows; 100sows/group) | | |
|---|---|---|---|
| | Control | GABA G100L100 | GABA G200L100 |
| Parity | 4.56±2.30 | 4.49±2.53 | 4.70±2.29 |
| GABA supplement (d) | 0 | 50.50±1.51$^{a}$ | 48.36±1.98$^{b}$ |
| Piglet born | 13.81±2.71 | 13.75±3.11 | 14.09±2.53 |
| Piglet born alive | 12.61±2.48 | 12.50±3.00 | 13.05±2.20 |
| Survival rate (%) | 91.31% | 90.91% | 92.62% |
| Litter birth weight (kg) | 18.94±3.94 | 19.35±4.55 | 19.82±3.32 |
| Average birth weight (kg) | 1.50±0.15$^{a}$ | 1.55±0.15$^{b}$ | 1.53±0.17$^{a}$ |
| Mummified fetus | 0.22±0.49 | 0.17±0.47 | 0.19±0.44 |

$^{a,b,c}$ $P<0.05$

| Details | Group (Total No=300sows; 100sows/group) | | |
|---|---|---|---|
| | Control | GABA G100L100 | GABA G200L100 |
| Sucking piglet/sow | 11.88±0.67$^{a}$ | 12.16±0.95$^{b}$ | 12.16±0.95$^{b}$ |
| Litter lactating weight (kg) | 17.88±1.20$^{a}$ | 18.87±2.10$^{b}$ | 18.57±2.56$^{b}$ |
| GABA supplement (d) | 0 | 26.33±1.71 | 23.91±1.22 |
| Lactating days | 24.79±1.55$^{a}$ | 26.33±1.72$^{b}$ | 23.91±1.23$^{c}$ |
| No. piglet wean/sow | 11.04±1.48$^{a}$ | 11.23±1.36$^{a}$ | 11.64±1.32$^{b}$ |
| Preweaning mortality (%) | 7.07% | 7.65% | 4.28% |
| Litter weaned weight (kg) | 71.04±14.71$^{a}$ | 77.50±16.04$^{b}$ | 75.47±12.00$^{b}$ |
| Average weaned weight (kg) | 6.41±0.99$^{a}$ | 6.88±1.11$^{b}$ | 6.39±0.87$^{c}$ |
| Weight gain (kg) | 4.90±1.01$^{a}$ | 5.33±1.11$^{b}$ | 4.97±0.89$^{a}$ |
| ADG (g/d) | 198.01±40.12 | 202.61±41.88 | 208.09±38.87 |
| WTS (d) | 5.01±1.74$^{a}$ | 6.17±4.62$^{b}$ | 5.32±3.40$^{a}$ |
| Total GABA supplement days | 0 | 76.83±1.75$^{a}$ | 72.27±2.01$^{b}$ |

# Take Home Messages

- GABA in animals illustrate their benefit beyond the gut
- The more studies of GABA, the more GABA benefit we found
- GABA can be a key ATA to reduce the AMU



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